SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	M.Deepika	**Reg. No.:**	192372296
Section / Batch:	2023	**Date:**	24/07/2026

Code of Conduct

I _____M.Deepika-192372296________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ___deepika.m____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25

Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
	GRAND TOTAL:	**100 Marks**	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, welldocumented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and welljustified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____________	Signature: _____________	Signature: _____________
Date: ________________	Date: ________________	Date: ________________

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7

Q1 **Binary Search** 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

Test Input

Very humid
Now

Search

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- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low <= high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}
```

Test Input

5 1

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7

Q2 **Stack** 5 marks

Find the error in the stack deletion operation. Justify it.
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int pop() {
    if(top == -1) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Test Input

stdin input (optional)

Output

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Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```
int pop() {
    if(top == -1) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks ✓
- Q2 Stack 5 marks ✓
- Q3 Stack 5 marks ✓
- Q4 Stack 5 marks ✓
- Q5 Stack - Balancing Symbols 5 marks ✓
- Q6 Singly Linked List 5 marks ✓

Q3 **Stack** 5 marks

Check the snippet and identify the logical error.
int stack[5];
int top;
void init() {
top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int stack[5];
int top;

void init() {
    top = -1;   // Correct initialization: stack starts empty
}

void push(int x) {
    if (top < 4) {          // Prevent overflow (stack[5] means valid
```

Test Input

stdin input (optional)

Output

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- Q6 Singly Linked List 5 marks ✓

- Q7 Stack 5 marks ✓
- Q8 Linked List 5 marks ✓
- Q9 Linked List 5 marks ✓
- Q10 Queue 5 marks ✓

10/10 answered

```
    top = -1;   // Correct initialization: stack starts empty
}

void push(int x) {
    if (top < 4) {          // Prevent overflow (stack[5] means valid
        top++;
        stack[top] = x;
    } else {
        printf("Stack Overflow\n");
    }
}

int pop() {
    if (top >= 0) {
        int val = stack[top];
        top--;
        return val;
    } else {
        printf("Stack Underflow\n");
        return -1;
    }
}

int isEmpty() {
    return top == -1;
```

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7

Q4 Stack

5 marks

Identify the errors in the following snippet.
if(ch == ')') {
while(stack[top] != '(') {
postfix[j++] = pop();
}
pop();
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(ch == ')') {
    while(top >= 0 && stack[top] != '(') {
        postfix[j++] = pop();
    }
    pop();  // pop the matching '(' as well
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == ')' && stack[top] == '(')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if((ch == ']' && stack[top] == '[') ||
   (ch == '}' && stack[top] == '{') ||
   (ch == ')' && stack[top] == '(')) {
    top--; // pop
} else {
    return 0; // mismatch
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7

Q6 **Singly Linked List** 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
temp = head
while temp is not None:
    print(temp.data)
    temp = temp.next
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7

Q7 **Stack** 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7

Q8 **Linked List** 5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
prev = None
current = head

while current is not None:
    next_node = current.next    # save next
    current.next = prev        # reverse link
    prev = current             # move prev forward
    current = next_node      # move current forward

return prev
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
DEQUEUE(Q):
    if front == -1:
        print("Underflow")
    else:
        x = Q[front]
        front = front + 1
        if front > rear:
            front = -1
            rear = -1
```

Test Input

stdin input (optional)

Output

Q5 Stack - Balancing Symbols 5 marks
Q6 Singly Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

10/10 answered

```
    if front == -1:
        print("Underflow")
    else:
        x = Q[front]
        front = front + 1
        if front > rear:
            front = -1
            rear = -1
        return x
```

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack 5 marks
Q2 Circular Queue 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Doubly Linked List 5 marks
Q6 Queue 5 marks
Q7

Q1 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
PEEK(stack):
return stack[top]
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7

Q2 Circular Queue 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if ((rear + 1) % MAX == front) {
    printf("Queue Full");
} else {
    rear = (rear + 1) % MAX;
    queue[rear] = vehicle;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7

Q3 Stack 5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
MAX = 5
stack = []
top = -1  # indicates empty stack

def push(value):
    global top
    if top == MAX - 1:
        print("Overflow")
    else:
        top += 1
        stack.append(value)
        print(f"Pushed {value}")
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7

Q4 Stack 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
POP():
if top == -1 → Underflow
else
    item = stack[top]
    top = top - 1
    return item
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7

Q5 Doubly Linked List 5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void deleteNode(Node* del) {
    if (del == NULL) return;

    if (del->prev != NULL)
        del->prev->next = del->next;
    else
        head = del->next;   // del was head

```

Test Input

stdin input (optional)

Output

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10

del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void deleteNode(Node* del) {
    if (del == NULL) return;

    if (del->prev != NULL)
        del->prev->next = del->next;
    else
        head = del->next;   // del was head

    if (del->next != NULL)
        del->next->prev = del->prev;
    else
        tail = del->prev;   // del was tail

    delete del;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7

Q6 **Queue** 5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void enqueue(int data) {
    if (rear == MAX - 1) return;   // Queue is full
    queue[++rear] = data;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7

Q7 Queue - Deque 5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void insertFront(int data) {
    if (front == -1 && rear == -1) {   // deque is empty
        front = rear = 0;
    }
    else if ((front - 1 + MAX) % MAX == rear) { // deque is full
        printf("Deque is full\n");
        return;
    }
    else {
```

Test Input

stdin input (optional)

Output

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Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7 Queue - Deque 5 marks

Q8 Binary Search 5 marks

Q9 Circular Queue 5 marks

Q10 Stack - Postfix Expression 5 marks

10/10 answered

Your Fixed Code

```
void insertFront(int data) {
    if (front == -1 && rear == -1) {   // deque is empty
        front = rear = 0;
    }
    else if ((front - 1 + MAX) % MAX == rear) { // deque is full
        printf("Deque is full\n");
        return;
    }
    else {
        front = (front - 1 + MAX) % MAX; // move front circularly back

    }

    deque[front] = data;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7

Q8 **Binary Search** 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int l, int r, int x) {
    if (l > r) return -1;

    int mid = l + (r - l) / 2;

    if (arr[mid] == x) return mid;

    if (arr[mid] > x)
```

Test Input

stdin input (optional)

Output

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- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

```
int binarySearch(int arr[], int l, int r, int x) {
    if (l > r) return -1;

    int mid = l + (r - l) / 2;

    if (arr[mid] == x) return mid;

    if (arr[mid] > x)
        return binarySearch(arr, l, mid - 1, x);

    return binarySearch(arr, mid + 1, r, x);
}
```

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7

Q9 **Circular Queue** 5 marks

Identify the bug in this Circular Queue dequeue logic.
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int dequeue() {
    if (front == -1) return -1;   // empty

    int data = queue[front];

    if (front == rear) {           // last element being removed
        front = -1;
        rear = -1;
```

Test Input

stdin input (optional)

Output

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- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

```
int dequeue() {
    if (front == -1) return -1;   // empty

    int data = queue[front];

    if (front == rear) {           // last element being removed
        front = -1;
        rear = -1;
    } else {
        front = (front + 1) % MAX;
    }

    return data;
}
```

stdin input (optional)

Output

Pending manual review

Submitted

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
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- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int i = 0;
while (expression[i] != '\0') {
    if (isdigit(expression[i])) {
        push(expression[i] - '0');
    } else {
```

Test Input

stdin input (optional)

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Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Your Fixed Code

```
int i = 0;
while (expression[i] != '\0') {
    if (isdigit(expression[i])) {
        push(expression[i] - '0');
    } else {
        int op2 = pop();
        int op1 = pop();
        int result;

        // perform operation (example for '+')
        result = op1 + op2;

        push(result);
    }
    i++;
}
```

Test Input

stdin input (optional)

Output

Pending manual review

Submitted

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